Unveiling Malware Traffic Indicators through DNS query Analysis: 2018-2024 Snapshot of Known Adversary Groups

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November 7, 2026

Abstract

The increasing sophistication of malware and its reliance on Domain Name System (DNS) communication presents a significant challenge to network security. While modern encryption methods have limited traditional analysis techniques, DNS remains an unencrypted protocol and a critical vector for malware operations, including command-and-control (C&C) communication via Domain Generation Algorithms (DGAs). This study addresses the need for proactive and scalable detection mechanisms by analyzing DNS query characteristics, such as TTL, response codes, entropy, and TLD patterns, over a six-year period. By focusing on the statistical behaviors of malware-induced DNS traffic, the research aims to unveil indicators that differentiate malicious activities from benign traffic, offering practical tools for cybersecurity operations and incident response.

The research analyzed DNS traffic data collected from two datasets: 1,000 malware-related PCAP files, resulting in 256,449 DNS queries, and 1 million benign DNS queries from Cisco Umbrella's popularity list. Key variables, such as TTL, entropy, and DNS query trends, were statistically analyzed to identify anomalies associated with malware activity. Information entropy calculations were applied to distinguish human-readable domain names from algorithmically generated ones, while temporal and frequency trends highlighted patterns unique to malware communication. The findings provide a basis for refining DNS-based malware detection techniques without relying on computationally expensive packet inspection.

Keywords: DNS, Malware, Domain Generation Algorithm, Botnet, Network Security

Research characteristics

Developmental Research: the research develops new methods for analyzing malware communication. Cross-Sectional Research: analyzing data at a specific point in time. examining DNS queries and TLD trends associated with known malware samples and comparing them against benign network traffic over the 2018-2024 timeframe.

1 Operational Definition & Problem Statement

1.1 DNS

Domain Name System (DNS) is a service that maps Internet websites and domain names to the corresponding IP addresses. DNS name resolution takes place before the user views a website and the attackers redirect or link users to other malicious website, with malware, drive-by-downloads, malicious scripts, malicious advertisements or other malicious content (**?**). the Round Trip Time of a query, in general, that is a good metric to estimate how quickly a server can provide an answer to a query (**?**).

The simple DNS query is generated at a system will go to a resolver. This resolver will inquire to different servers for the allocated IP address, and the response will be sent back to your computer. DNS doesn't work on just a single server; instead, a large number of servers are hierarchically, and these servers are spread over the world.

The root DNS server, TLD, and authoritative DNS are the three basic servers where TLD servers consist of generic domain names like gov, edu, com, org. TLD server gives the record to the corresponding authoritative DNS server, which maintains the domain. Finally, the authoritative DNS server returns the IP address of the website (**?**).

1.1.1 Response Code

RCODE Response code - this 4 bit field is set as part of responses. The values have the following interpretation (**?**):

0 No error condition

1 Format error - The name server was unable to interpret the query.

2 Server failure - The name server was unable to process this query due to a problem with the name server.

3 Name Error - Meaningful only for responses from an authoritative name server, this code signifies that the domain name referenced in the query does not exist.

4 Not Implemented - The name server does not support the requested kind of query.

Figure 1: Sample DNS request journy

5 Refused - The name server refuses to perform the specified operation for policy reasons.

1.1.2 TTL

Time To Live (TTL) controls how long a specific information can be maintained in the cache of a resolver, in this research we meant to use TTL for DNS Answers. A higher TTL value generally provides a better cache hit-rate, as it is not necessary to constantly replace the information (**?**).

Time-To-Live values (TTLs) of DNS records control cache durations and, therefore, affect latency, resilience, and the role of DNS in CDN server selection (**?**). However, if the TTL is too large, the probability of having stale information in the cache is higher. Therefore, a key point of contention for optimization of resolvers is proper management of its cache, by the way of tuning the TTL parameter (**?**).

TTLs in use range from as short as 5 minutes, to a few hours, to one or two days. This wide range of time values seen in TTL configurations is because there are many trade-offs in "short" vs. "long", and which factors are most important is specific to each organization (**?**).

1.1.3 FQDN & TLD

A complete domain name for a node is called a Fully Qualified Domain Name (FQDN). An FQDN defines a complete path for a domain name starting on the leaf (the host name) all the way to the root of the tree. Each node in the tree has its label that defines the zone. An example of an FQDN is "www.auckland.ac.nz.". A domain is a subdomain when it is contained in another domain; in the previous example "auckland.ac.nz" is a subdomain of "ac.nz" (**?**).

In the Domain Name System (DNS) naming of computers there is a hierarchy of names. The root of system is unnamed. There are a set of what are called "top-level domain names" (TLDs). These are the generic TLDs (EDU, COM, NET, ORG, GOV, MIL, and INT), and the two letter country codes from ISO-3166. It is extremely unlikely that any other TLDs will be created (**?**).

1.2 Malware & Botnet

The usage of DNS by botnet operators and malware writers poses significant challenges to detection and mitigation. Among the consequences of this usage is the utilization of a covert channel that seems particularly effective in achieving the criminal goals of the malware. This is, in part, due to the typical lack of filtration of DNS traffic.

Botnets are one of the the main attack vectors on the internet today. They are the root cause of many malicious activities in computer networks, such as denial of service, spam distribution, click fraud, adware, distributed brute-forcing of remote services, identity and data theft and many more.

A typical botnet consists of a number of malware-compromised machines, called bots, that are remotely controlled by a botmaster using a command and control (C&C) channel. Exploitation of a machine starts with a malware infection from a malicious web page, email attachment, etc. As soon as the malware infects a host, it usually tries to establish a connection to one or more C&C servers to download updates and retrieve commands or send private information gained from the host (**?**).

Software that "deliberately fulfills the harmful intent of an attacker" is referred to as malicious software or malware. These are intended to gain access to computer systems and network resources, disturb computer operations, and gather personal information without taking the consent of system's owner, thus creating a menace to the availability of the internet, integrity of its hosts, and the privacy of its users.

Malwares come in wide range of variations like Virus, Worm, Trojan-horse, Rootkit, Backdoor, Botnet, Spyware, Adware etc. These classes of malwares are not mutually exclusive meaning thereby that a particular malware may reveal the characteristics of multiple classes at the same time (**?**).

1.2.1 DGA

Malware is known to use DGA (Domain Generation Algorithms) to maintain its communication channels. A DGA algorithm is a resilient method to generate domains that allows the attacker to maintain control of the malware by forbidding the defenders to know which domain will be used next.

Detecting DGA traffic is very important to identify the current infected computers in the network. Detecting the domain names generated by a DGA is difficult because they are usually randomly generated from letters or common dictionary words. However we hypothesize that the behavior in the network generated by DGA is quite different from normal DNS traffic (**?**).

1.3 Information entropy

Information entropy is a measure of the variation of the components that make up a message. Entropy, as applied here, is calculated as the probability of a particular byte occurrence p(xi) multiplied by the logarithm of the probability of that occurrence, summed up for all byte occurrences. The mathematical formula is depicted below (**?**):

$$H(X) = -\sum p(x) \log_2 p(x)$$

where $p(x)$ represents the probability of each character in the domain string. Higher entropy values are indicative of DGA-generated domains, while lower val-

ues suggest human-readable domains. Temporal entropy trends were analyzed to identify periods of increased DGA usage.

The hypothesis behind the use of entropy is that the request queries for a particular website flow will produce a specific entropy distribution.

2 Research Necessity

Nowadays, almost all network protocols have become encrypted, decryption of this volume of data (if even possible), and then analyzing it against known signatures, cannot be effective.

On the other side, we have DNS protocol that is not encypted, and somehow necessary to exists even in malware programs, because having and remembering one static IP address, would not be possible in many aspects. Even when analyzing C&C Server communications, DNS traffic become the main context of malware network data.

The problem here is, DNS itself doesn't provide lots of valuable information to determine this data can be from a malware program or not, instead give us some time or age related varibales. We're going to use this special and numerical maybe even string variables, to statistically analyzing them, hope this can light up our way.

in this research, we would try to find a distinguished range of variables values in the datasets, in other word as the title of research claims that, finally some indicators will be introduced. these indicators later can be examined and fully tuned in different situations.

3 Research Background

1. Botnets have emerged as a major threat to modern computing due to their capacity for enabling diverse criminal activities, including spam dissemination, distributed denial-of-service (DDoS) attacks, and data theft. Hands, Yang, and Hansen (2015) (**?**) investigated the sophisticated communication methods utilized by botnets, particularly focusing on the role of the Domain Name System (DNS). Their study highlighted how botmasters exploit DNS for covert communication and resilience, employing techniques such as Domain Generation Algorithms (DGAs) and DNS tunneling. These approaches allow botnets to maintain control and evade detection through frequent updates and obfuscation of malicious activities. The researchers emphasized that proactive detection methods, such as entropy-based statistical analysis and network flow monitoring, are essential for identifying and mitigating botnet operations that leverage DNS channels. This foundational work underscores the importance of understanding DNS-based malware behavior for developing robust security measures.

2. In the domain of DNS-based malware detection, Grill et al. (2015) (**?**) made significant contributions by developing a novel approach to identifying Domain Generation Algorithm (DGA) malware using only NetFlow in-

formation. Their research demonstrated that DGA-based malware could be detected by analyzing the statistical relationship between DNS requests and visited IP addresses, without requiring deep packet inspection or reverse engineering. By modeling the ratio between DNS queries and unique IP connections, they achieved detection rates with AUC values exceeding 0.97 across multiple experiments. This work was particularly innovative because it showed that privacy-preserving network monitoring could effectively identify malicious activities while minimizing computational overhead and avoiding the need for packet-level analysis. Their findings align with our research direction, as they similarly focus on leveraging DNS query patterns to identify malicious activities, though our approach expands upon this by incorporating additional parameters such as temporal entropy analysis and TLD preferences to provide a more comprehensive detection framework.

3. Building upon groundbreaking research by Matthew Thomas and Aziz Mohaisen from Verisign Labs in 2014 (**?**), the detection and analysis of malware command and control (C2) infrastructure through DNS traffic analysis has emerged as a crucial aspect of modern cybersecurity. Their pioneering study established that malware utilizing Domain Generation Algorithms (DGAs) produces distinctive DNS lookup patterns that can be identified through traffic similarity metrics at the authoritative name server level. DGAs, which generate pseudo-random domain names for C2 communication, became prevalent following their successful deployment by the Conficker malware family, leading to a proliferation of similar techniques across various threat actors. While malware operators typically register only a small subset of generated domains for actual C2 operations, the remaining unregistered domains generate significant Non-Existent Domain (NXDomain) traffic that exhibits unique patterns distinguishable from legitimate traffic. These patterns, characterized by concurrent queries from numerous recursive name servers across different geographic locations, create identifiable traffic similarity signatures that have formed the foundation for modern malware detection techniques. This historical context provides essential groundwork for analyzing the evolution of DNS query behavior associated with known adversary groups from 2018 to 2024, as threat actors continue to adapt their techniques to evade detection while maintaining necessary command and control infrastructure.

4. Previous research by Hao, Feamster, and Pandrangi, published in 2011, (**?**) provides important foundational work for studying malicious domain behavior through DNS analysis. Their study examined newly registered domains under .com and .net TLDs, analyzing both their DNS infrastructure characteristics and lookup patterns from recursive resolvers. By monitoring DNS resource records and query traffic at top-level domain servers, they discovered several key patterns distinguishing malicious from legitimate domains. Their findings revealed that about 55% of malicious domains were used in attacks at least one day after registration, suggesting potential for early detection. They also identified specific IP address ranges and "tainted" Autonomous Systems that predominantly hosted malicious DNS infrastructure. Additionally, their research showed that malicious domains exhibited distinct lookup patterns, receiving

queries from a more diverse set of networks and experiencing rapid spikes in lookup volume shortly after registration. These patterns were often evident before attacks occurred, indicating potential for developing early warning systems for malicious domain detection.

4 Methodology, Tools and Techniques

4.1 Pupolation and Data gathering

4.1.1 Malware Dataset

We have gathered our malware traffic data from malware-traffic-analysis.net (**?**), 1000 Malware Traffic record PCAP file has been gatherd at this stage, later these files became 256449 DNS Queries.

The website is well-regarded within the cybersecurity community. Users praised it as a valuable and safe resource for analyzing pcap files, noting its utility in professional exercises and training. its frequent mentions in cybersecurity discussions and resources suggest it is a respected and utilized platform among professionals and enthusiasts in the field.

4.1.2 Normal Dataset

For the Normal dataset, We have got 1 Million Most-Queried FQDNs from Cisco Umbrella Popularity List (**?**).

The popularity list contains most queried domains based on passive DNS usage across Umbrella global network of more than 100 Billion requests per day with 65 million unique active users, in more than 165 countries. the metric is not based on only browser based 'http' requests from users but rather takes into account the number of unique client IPs invoking this domain relative to the sum of all requests to all domains. In other words, popularity ranking reflects the domain's relative internet activity agnostic to the invocation protocols and applications (**?**). It is commonly used for Baseline Analysis, Threat Research and Network Monitoring.

4.2 Data Preprocessing

4.2.1 Zeek - Network Data Processing

PCAP files are large binary data files that have lots of additional network stack layers info. To process such data, we have make it in the form of well-structured data from DNS Layer and its Queries (Ex. JSON).

Zeek is open-source NIDS tool and its architecture is highly scalable in which performance improvements can be easily achieved by dedicating more hardware resources. Zeek coined its importance in the environments where zero-day attacks are concerned as it also supports anomaly-based detection. Zeek provides an in-depth analysis of network traffic, and it is quite an efficient, easily deployable, and flexible open-source solution. (**?**)

Figure 2: Before Data Cleaning

Using Zeek tool to gather JSON summary of DNS transactions in the Malware PACP files, for the Normal Dataset, since we have just the query itself not the full DNS packet or PCAP file, we would use "dig" command in linux to make the DNS requests in real-time, and capturing it using Zeek tool in NIC (Network Interface) listen mode, finally gathering JSON as output.

In this research, we also incorporate "icannTLD" (**?**) Zeek plugins to Extract and separate TLD, Domain, Sub-Domain from the Query string variable. Due to ccTLD issue that has arisen from Zeek TLD extract plugin, "tldextract" (**?**) python library help us in fixing the issue and have declare which query is ccTLD or not, by introducing new boolean variable.

Entropy calculation of Queries has been done using Scripting capabilities of the tool itself, changes were made in DNS processing script to make this capability enable, and finally add a numeric variable that store the value.

4.2.2 PySpark - Python JSON Processing

Apache Spark as the PySpark Python Library would ingest the JSON file gathered from python script in the last step, add year column to malware dataset, and a column to be the indicator of being malware or not in all of records from the two datasets (in normal dataset "0" or "2025" can be used as the year).

As the next-generation engine for big data analytics, Apache Spark can alleviate key challenges of data preprocessing, iterative algorithms, interactive analytics and operational analytics among others. Since its initial releases, Apache Spark has seen a rapid adoption by enterprises across a wide range of industries.

Such fast adoption with the potential of Apache Spark as a unified processing engine, which integrates with many storage systems, has led to dozens of community-contributed packages that work with Apache Spark. Apache Spark has been leveraged as a core engine in many world-class companies such as IBM, Huawei, Tencent and Yahoo (**?**).

4.3 Implementation limitations

The research use Top 1 Million Queries but this is not exactly normal DNS traffic, due to being prone to false positives and unstable environment along with Ethical issues that can arise from network connection capturing in public environment, we were unable to collect more ordinary dataset. So we had to produce it.

Malware Dataset of course doesn't consist of all the possible and available malwares in the wild, but the website (**?**) have a lot known and released malware.

Figure 3: After Data Cleaning - TTL analysis

4.4 Data cleaning and Analysis

As this malware dataset gathered from live laboratory Windows systems, has many ordinary queries that should exist in a windows network environment. To fix this we will use PySpark to delete any of queries or domains that is same in the two datasets. Some queries may have Blank TLD, or have unoffical form of FQDN, those must be removed too.

After data cleaning, we will divide the DataFrame into three different DataFrames for TTL analysis, RTT analysis, and entropy analysis, each with its unique combination of columns / variables to be assessed in future statistical test.

To have same weight for each domain, Parameters all of queries that have same domain name must be in a list, final DataFrame should be in form of "Domain" - "List of parameters" pairs. Then, we can remove duplicate rows (same domain name) to obtain unique rows. We need to follow the same process for the normal DataFrame as well. Now, we can perform statistical analysis on these parameters lists.

Parameters include (but not limited to): Entropy of Query String, Time To Live (TTLs), Round Trip Time (RTT). We need to follow the same process for the other DataFrames as well.

5 Ethical issues

In this research normal dataset has been produced and captured in laboratory conditions, and we have used publicly available Google DNS servers (**?**), so no Ethical issues can exists.

Malware dataset as Bradley Duncan (**?**) claims, their malware tests captured in laboratory conditions, so no Ethical issues can exists.

6 Objective & Hypothesis

6.1 Objective

6.1.1 General objective

Finding a range of parameters that can give us a classification criterion for Separating normal DNS traffic from malware DNS traffic

6.1.2 Specific objectives

1. Measuring and comparing TTL values in the normal dataset and malware dataset after data cleaning.

2. Measuring and comparing RTT values in the normal dataset and malware dataset after data cleaning.

3. Measuring and comparing Entropy values in the normal dataset and malware dataset after data cleaning.

4. Measuring and comparing TLD distribution in the normal dataset and malware dataset after data cleaning.

6.1.3 Practical purposes

If the parameters have Significant difference, we would use the median range (Between Q1 of first dataset and Q3 of second one) of the malware and normal datasets for classification of future DNS queries.

6.2 Hypothesis

1. TTL values in the two datasets have Significant difference.
2. RTT values in the two datasets have Significant difference.
3. Entropy values in the two datasets have Significant difference.
4. TLD distribution in the two datasets have Significant difference.
5. Rcode values in the two datasets have difference.
6. Length values in the two datasets have difference.

7 Thesis schedule

Data Collection : 3 weeks
Statistical Analysis : 2 weeks
Thesis Writing : 2 weeks